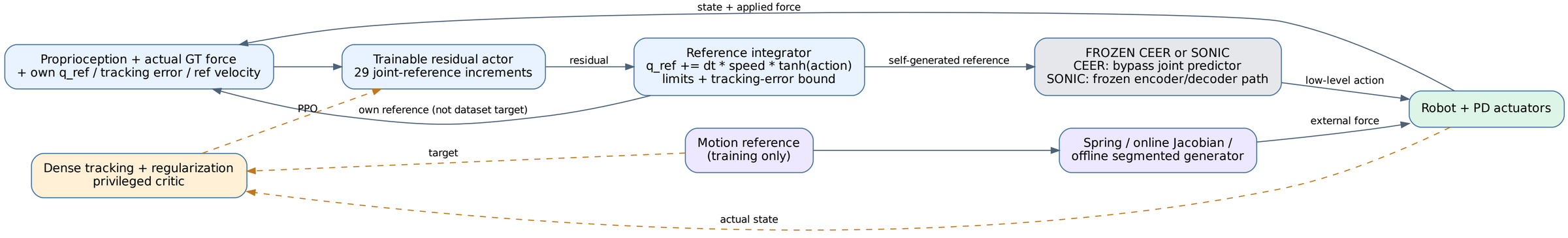
C | GT-force residual actor over a frozen motion tracker
state + applied force
Proprioception + actual GT force
+ own q_ref / tracking error / ref velocity
Trainable residual actor
29 joint-reference increments
residual
Reference integrator
q_ref += dt * speed * tanh(action)
limits + tracking-error bound
self-generated reference
FROZEN CEER or SONIC
CEER: bypass joint predictor
SONIC: frozen encoder/decoder path
low-level action
Robot + PD actuators
PPO
own reference (not dataset target)
external force
Motion reference
(training only)
Spring / online Jacobian /
offline segmented generator
target
Dense tracking + regularization
privileged critic
actual state